System Requirement Specification Document

1. Introduction

- **Purpose:**

The purpose of this System Requirement Specification (SRS) document is to provide a detailed and comprehensive description of the Salary Management System designed for a government organization. This document aims to capture all the necessary requirements for the development, implementation, and maintenance of the system. It is intended for use by the development team, project managers, stakeholders, and any future maintainers of the system.

- **Scope:**

The Salary Management System is designed to streamline the payroll process for the public administration department. It includes functionalities for managing employee salaries, including overtime payments. The system aims to reduce errors, save administrative time, and provide a secure and efficient means of handling payroll data. Only authorized personnel, specifically the secretary of the relevant branch, can access the system. The key functionalities include:

- Managing employee details.

- Handling overtime (OT) details.

- Searching for employee information.

- Authenticating user through a secure login form.

- Navigating between different forms via a home dashboard.

- **Definitions, Acronyms, and Abbreviations:**

- SRS: System Requirement Specification

- OT: Overtime

- SQL: Structured Query Language

- UML: Unified Modeling Language

- C#: A programming language developed by Microsoft

- MySQL: An open-source relational database management system

- **References:**

  - [IEEE Standard 830-1998](https://ieeexplore.ieee.org/document/720574): IEEE Recommended Practice for Software Requirements Specifications

  - MySQL Workbench Documentation

  - Microsoft C# Documentation

This SRS document serves as a foundation for the development of the Salary Management System and ensures that all requirements are clearly understood and agreed upon by all parties involved.

2. Description

- **Product Perspective:**

The Salary Management System is a part of the government's payroll management infrastructure. It uses a MySQL database for storing and managing employee and salary data and ensures secure user authentication for authorized access.

- **Product Functions:**

- User Authentication: Secure login for authorized access.

- Employee Management: Add, edit, and view employee details.

- Overtime Management: Record and update overtime hours and payments.

- Search Functionality: Search for employee details using their ID.

- Dashboard Navigation: Main dashboard for easy access to different system functions.

- **User Characteristics:**

The system is designed for administrative staff, particularly secretaries in government branches, with basic computer skills and familiarity with forms and databases.

- **Constraints:**

- Regulatory Requirements: Must comply with data privacy and security regulations.

- Hardware Limitations: Should run on standard office computers.

- Security: Strong authentication to protect payroll data.

3. **Specific requirements:**

- **Functional Requirements:**

  - User Authentication: Validate user credentials to grant access.

  - Employee Management: Add, edit, and view employee details.

  - Overtime Management: Record and update overtime hours and payments.

  - Search Functionality: Search for employees by their ID and display details.

- **Non-functional Requirements:**

  - Performance: The system should handle up to 100 simultaneous users without performance degradation.

  - Security: Passwords must be encrypted, and user sessions should be secured.

  - Usability: The interface should be intuitive and easy to navigate for users with basic computer skills.

  - Reliability: The system should have a 99.9% uptime, ensuring availability during office hours.

- **External Interface Requirements:**

  - Database Interaction: Connect to the MySQL database to fetch and store employee and salary details.

  - APIs: Future integration with other government payroll systems using standard APIs.

- **System Features**

1. Login System:

- Inputs: Username, password.

- Processing: Verify credentials against the users table.

- Outputs: Access to the system if credentials are correct; error message if incorrect.

2. Employee Details Management:

- Inputs: Employee details (name, position, address, etc.).

- Processing: Add or update employee information in the database.

- Outputs: Confirmation message; updated Data Grid View with employee details.

3. Overtime Details Management:

- Inputs: Overtime hours and payment details.

- Processing: Record or update overtime information in the database.

- Outputs: Confirmation message; updated Data Grid View with overtime details.

4. Search Employee:

- Inputs: Employee ID.

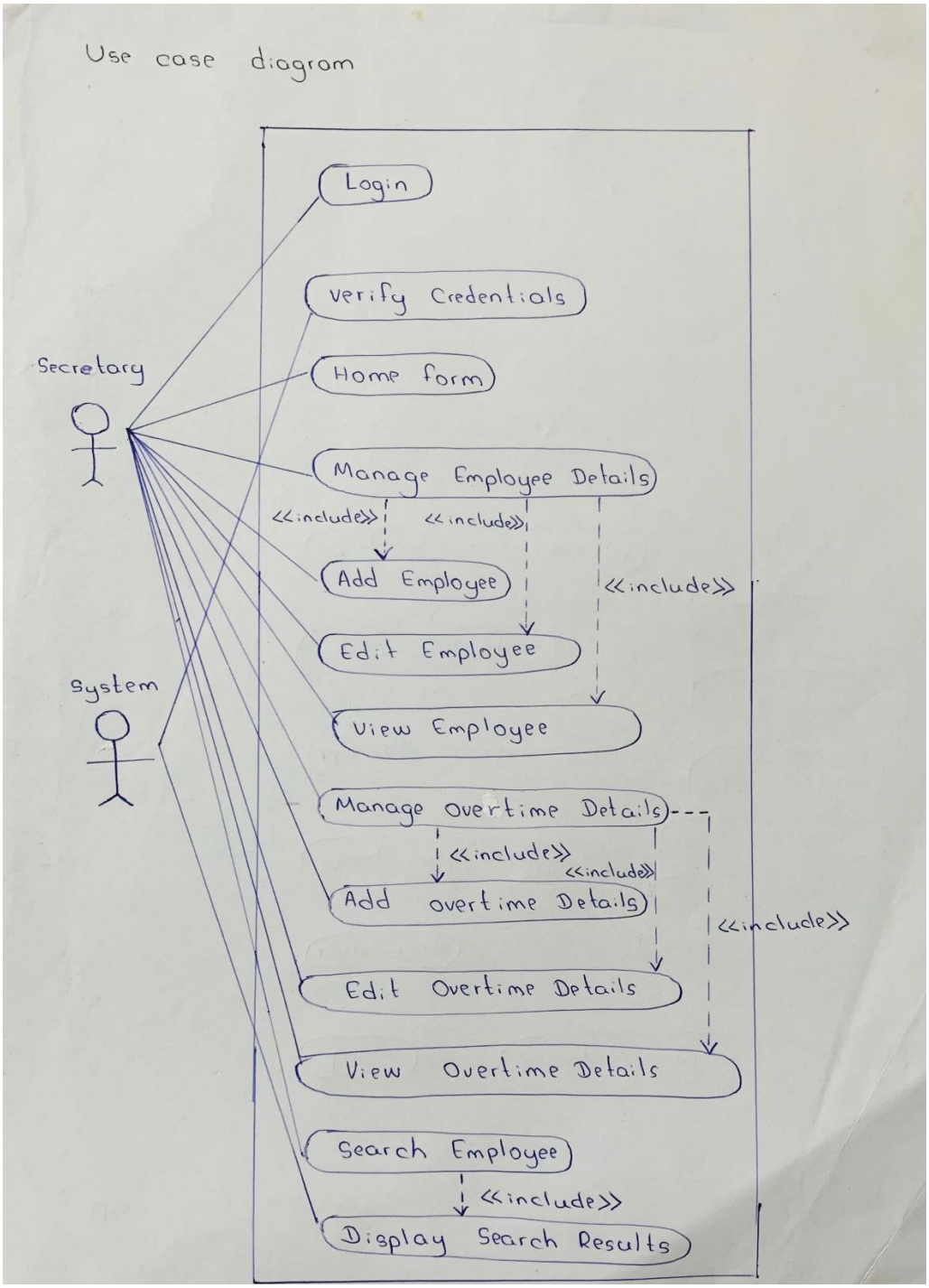
- Processing: Retrieve employee details from the database.

- Outputs: Display employee information in textboxes; error message if no record found.

4. System Models

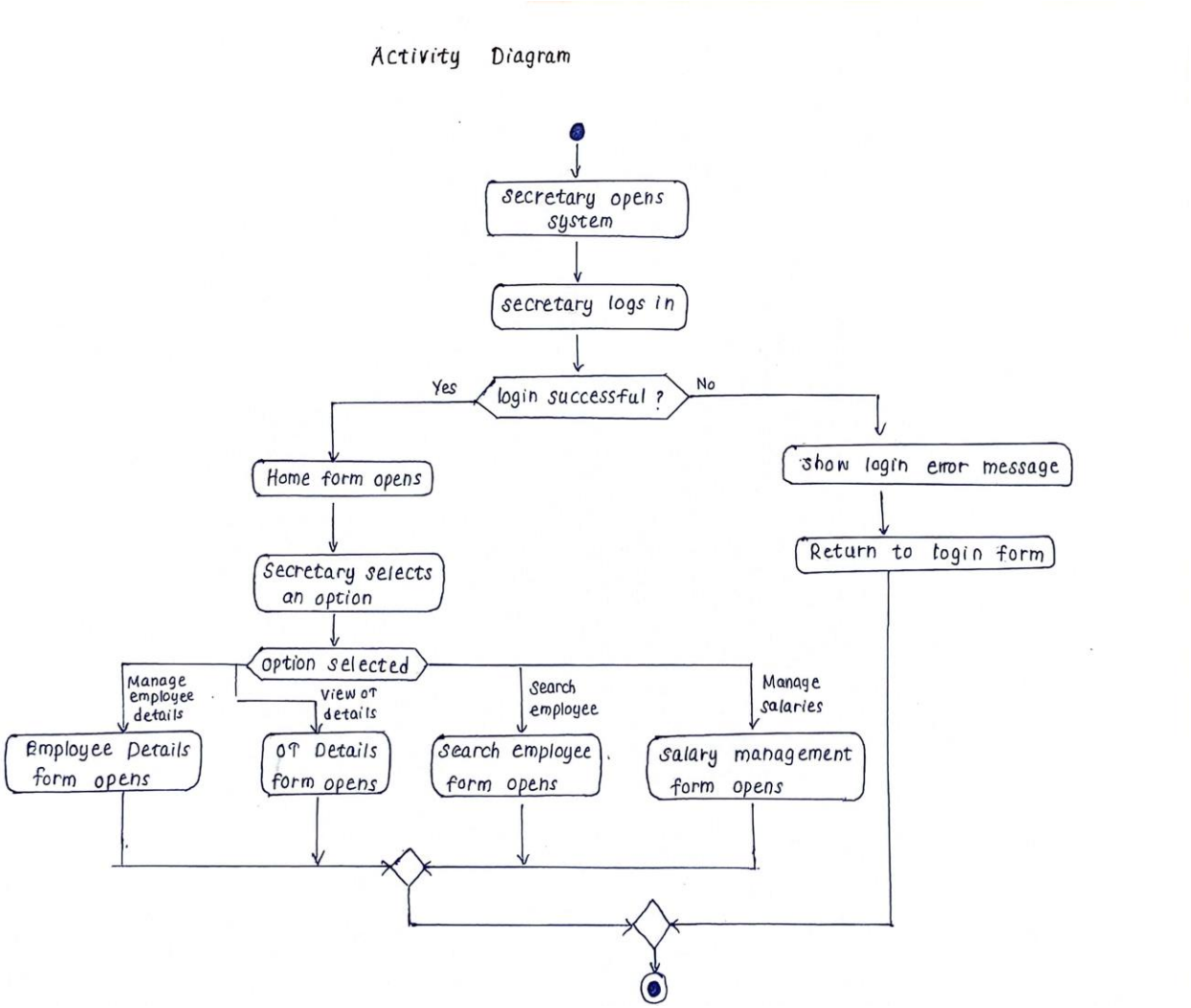
- **Use Case Diagrams**

  This diagram shows the main interactions between the users and the system. The primary actors are the Secretary and the System.



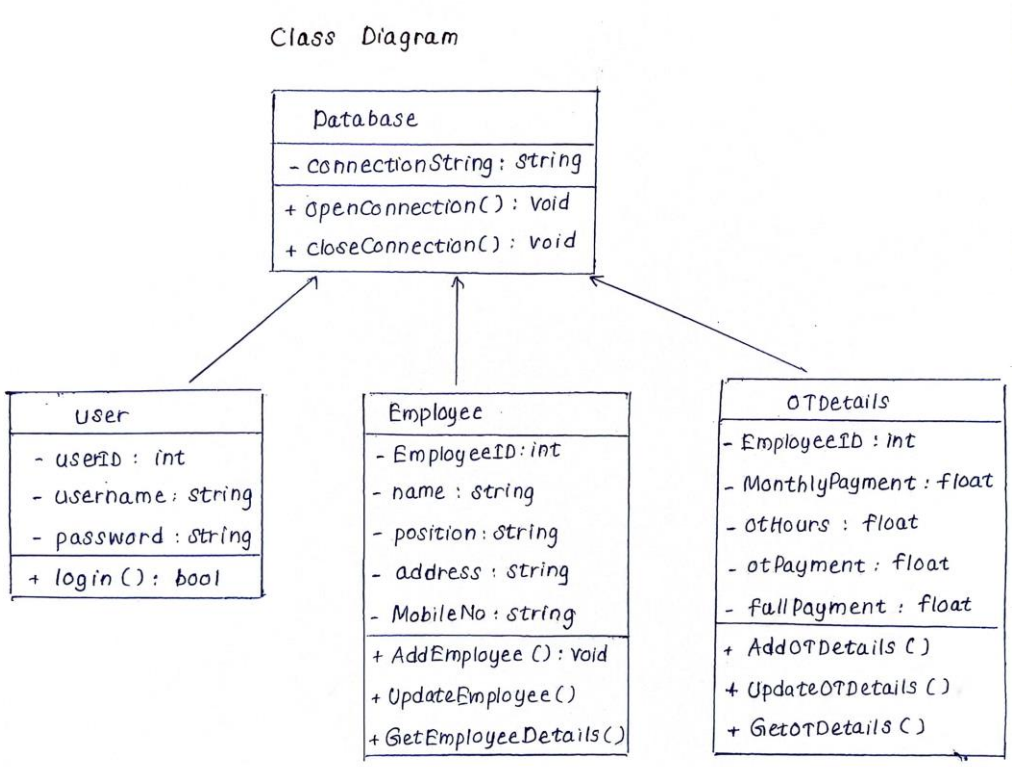
- **Activity Diagrams**

  This diagram illustrates the workflow of the user authentication process in the Salary Management System.



- **Class Diagrams**

  This diagram illustrates the main classes, their attributes, and relationships in the Salary Management System.



5. Appendices

- **Glossary:**

  Terms used in the document:

  - **SRS**: System Requirement Specification
  - **OT**: Overtime
  - **SQL**: Structured Query Language
  - **UML**: Unified Modeling Language

- **List of Figures**

  - **Use Case Diagram**: Visual representation of user interactions with the system.
  - **Activity Diagram**: Workflow of the login process.
  - **Class Diagram**: Structure of the system, showing classes, attributes, and relationships.

- **References**

  - **MySQL Documentation**: Official documentation for MySQL database.
  - **C# Programming Guide**: Microsoft's guide for C# programming.
  - **UML Diagrams**: Resources and examples for creating UML diagrams.